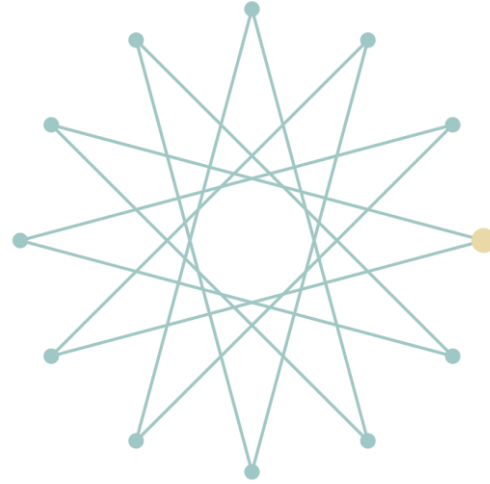




Identity-Based Digital Signatures: the Guillou-Quisquater Scheme (in $\mathbb{Z}_n$)

Modern Cryptographic Techniques

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Introduction



Traditional PKI vs. IBC

Identity-Based Cryptography (IBC)

A public identifier — an email address, a name — be used directly as a public key.

The Guillou-Quisquater Scheme (1988)

An identification and digital signature protocol built on the hardness of the RSA problem.

Main Advantage

Removes the need for public-key certificates — a simplified PKI.

Traditional PKI

User → generates keys → requests a certificate → CA verifies → issues certificate → distributes certificate

IBC (Guillou-Quisquater)

User → submits ID → CA computes a secret certificate → done!

Public Key = Identity

alice@company.com → public key

Theoretical Foundations

$$\mathbb{Z}_n^* = \{ a \in \mathbb{Z}_n \mid \gcd(a, n) = 1 \}$$

The set of numbers in $[1, n-1]$ that share no common factor with n

$\gcd(a, n)$ = greatest common divisor

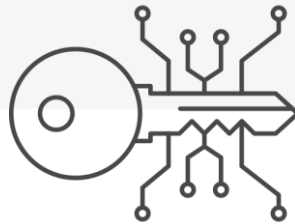
The largest positive integer that divides both a and n

Example: $n = 15$

$\mathbb{Z}^*_{15} = \{1, 2, 4, 7, 8, 11, 13, 14\}$

For $n = p \times q$ (primes)

Group order: $|\mathbb{Z}^*_n| = \varphi(n) = (p-1)(q-1)$



Key properties for GQ:

$a \times b \bmod n$

Modular multiplication — a closed group

$a^k \bmod n$

Modular exponentiation — efficient (square-and-multiply)

$a^{-1} \bmod n$

Modular inverse — exists for every element of $\mathbb{Z}^*_n$

$a^{1/v} \bmod n$

v -th root — HARD without factoring n !

The GQ Scheme: Setup & Certification

STEP 1: System Setup (CA)

Generate large primes:

$p, q \leftarrow \text{large primes (512-1024 bits)}$

Compute:

$$n = p \times q$$

$$\phi(n) = (p-1)(q-1)$$

Choose the public exponent:

$$v = 65537$$

Compute the secret exponent:

$$s = v^{-1} \bmod \phi(n)$$

Public: n, v, k | **Secret:** p, q, s

STEP 2: Certificate Generation

For a user with identity ID (e.g., alice@company.com):

1. Hash the identity:

$$I = H(\text{ID}) \bmod n$$

2. Compute the secret certificate:

$$J = I^s \bmod n$$

Key property: $J^v \equiv I \pmod{n}$

Proof: $J^v = I^{sv} = I^{1+k\phi(n)} = I \times 1^k = I$

The GQ Scheme: Signing & Verification

STEP 3: Signing (User)

- 1 Choose a random r from $\mathbb{Z}^*n$
 $r \leftarrow \text{random}, \gcd(r,n)=1$
 $\gcd(r,n)=1$
- 2 Compute the commitment
 $T = r^v \bmod n$
- 3 Compute the challenge
 $d = H(M \parallel T) \bmod 2^k$
- 4 Compute the response
 $y = r \times J^d \bmod n$

Signature: $\sigma = (d, y, ID)$

STEP 4: Verification (Anyone)

- 1 Recompute I from the identity
 $I = H(ID) \bmod n$
- 2 Reconstruct the commitment
 $T' = y^v \times I^{-d} \bmod n$
- 3 Recompute the challenge
 $d' = H(M \parallel T')$
- ⌚ Check equality
ACCEPT if $d' = d$

Correctness:

$$T' = y^v \times I^{-d} = r^v \times J^{vd} \times I^{-d} = r^v \times I^d \times I^{-d} = r^v = T$$

Java Implementation

Project Structure

GQSignature.java

CA constructor: generates p , q , n , v , s

User constructor: receives n , v , k

Methods: generateCertificate, sign, verify

GQDemo.java

Walkthrough with Alice and Bob

Integrity and authenticity tests

GQBenchmark.java

GQ vs RSA comparison, 100 iterations



Operations in $\mathbb{Z}^*_n$ (BigInteger)

Modular multiplication

`a.multiply(b).mod(n)`

Modular exponentiation

`a.modPow(e, n)`

Modular inverse

`a.modInverse(n)`

Prime generation

`BigInteger.probablePrime(bits, rnd)`

Conclusions

Advantages of the GQ Scheme

Removes the need for traditional PKI
Identity = public key (intuitive)
Faster signing for larger keys

Limitations

Key escrow: the CA knows every secret
Larger signatures (+25% bytes)
More complex identity revocation

Implementation Summary

- Complete implementation in Java
- Validated with integrity and authenticity tests
- Performance competitive with RSA



References

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